Experiment 25

MySQL Queries for QR-Enabled Automatic Bus Ticket Booking System

Aim

To manage passengers, bus schedules, bookings, fare calculations, booking status, and QR ticket references using MySQL.

Software Required

MySQL Workbench or MySQL Command Line Client.

SQL Program

Step 1: Create database

```
CREATE DATABASE BusBookingDB;
USE BusBookingDB;
```

Step 2: Create tables

```
CREATE TABLE Passenger (
    passenger_id INT PRIMARY KEY,
    passenger_name VARCHAR(50),
    phone VARCHAR(15)
);

CREATE TABLE Bus (
    bus_id INT PRIMARY KEY,
    bus_name VARCHAR(50),
    source VARCHAR(50),
    destination VARCHAR(50),
    fare DECIMAL(8,2),
    available_seats INT
);

CREATE TABLE Booking (
    ticket_id INT PRIMARY KEY,
    passenger_id INT,
    bus_id INT,
    journey_date DATE,
    seats INT,
    total_fare DECIMAL(10,2),
    booking_status VARCHAR(20),
    qr_code VARCHAR(100),
    FOREIGN KEY (passenger_id) REFERENCES Passenger(passenger_id),
    FOREIGN KEY (bus_id) REFERENCES Bus(bus_id)
);
```

Step 3: Insert sample records

```
INSERT INTO Passenger VALUES
(1, 'Ravi', '9876543210'),
(2, 'Priya', '9876501234'),
(3, 'Arun', '9876512345');

INSERT INTO Bus VALUES
(101, 'Express Bus', 'Chennai', 'Bangalore', 500, 40),
```

```
(102, 'City Travels', 'Chennai', 'Coimbatore', 400, 30),
(103, 'Super Deluxe', 'Bangalore', 'Hyderabad', 700, 25);

INSERT INTO Booking VALUES
(1001, 1, 101, '2026-11-05', 2, 1000, 'Confirmed', 'BUS-TICKET-1001'),
(1002, 2, 102, '2026-11-06', 1, 400, 'Confirmed', 'BUS-TICKET-1002'),
(1003, 3, 103, '2026-11-07', 1, 700, 'Pending', 'BUS-TICKET-1003');
```

Queries

Display buses

```
SELECT * FROM Bus;
```

Search a route

```
SELECT * FROM Bus WHERE source = 'Chennai' AND destination = 'Bangalore';
```

Display booking details

```
SELECT p.passenger_name, b.bus_name, b.source, b.destination,
       bk.journey_date, bk.seats, bk.total_fare, bk.booking_status
FROM Passenger p
JOIN Booking bk ON p.passenger_id = bk.passenger_id
JOIN Bus b ON bk.bus_id = b.bus_id;
```

Calculate fares

```
UPDATE Booking bk
JOIN Bus b ON bk.bus_id = b.bus_id
SET bk.total_fare = bk.seats * b.fare;
```

Create QR reference

```
UPDATE Booking SET qr_code = CONCAT('BUS-TICKET-', ticket_id) WHERE ticket_id = 1001;
```

Confirm booking

```
UPDATE Booking SET booking_status = 'Confirmed' WHERE ticket_id = 1003;
```

Display confirmed tickets

```
SELECT * FROM Booking WHERE booking_status = 'Confirmed';
```

Count confirmed bookings

```
SELECT COUNT(*) AS confirmed_tickets FROM Booking WHERE booking_status = 'Confirmed';
```

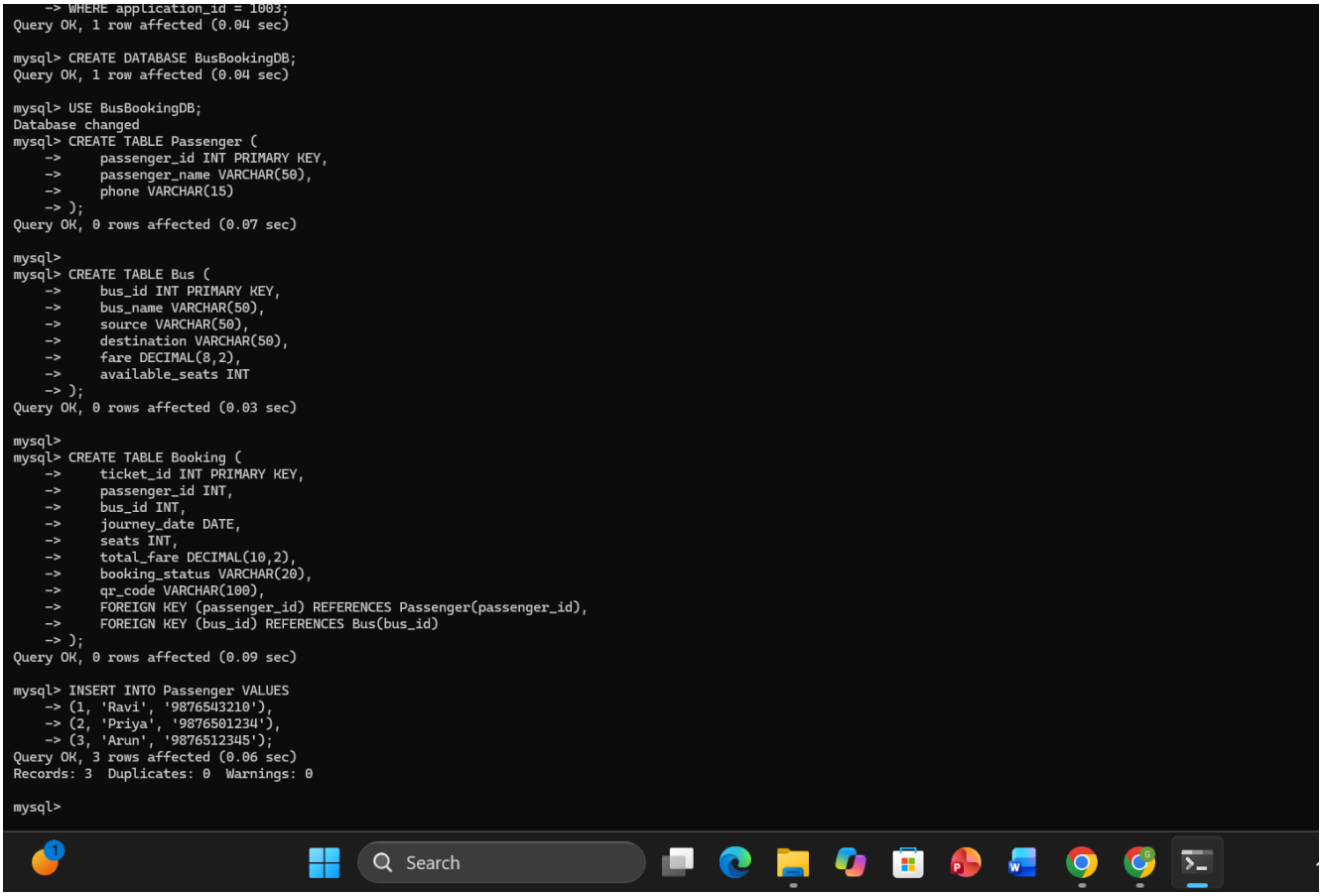
Display ticket QR information

```
SELECT p.passenger_name, b.bus_name, bk.ticket_id, bk.journey_date,
       bk.total_fare, bk.qr_code
FROM Passenger p
JOIN Booking bk ON p.passenger_id = bk.passenger_id
JOIN Bus b ON bk.bus_id = b.bus_id
WHERE bk.booking_status = 'Confirmed';
```

Verify QR reference

```
SELECT ticket_id, passenger_id, bus_id, journey_date, booking_status
FROM Booking
WHERE qr_code = 'BUS-TICKET-1001'
AND booking_status = 'Confirmed';
```

Expected Output:



```
mysql>
mysql> INSERT INTO Bus VALUES
    -> (101, 'Express Bus', 'Chennai', 'Bangalore', 500, 40),
    -> (102, 'City Travels', 'Chennai', 'Coimbatore', 400, 30),
    -> (103, 'Super Deluxe', 'Bangalore', 'Hyderabad', 700, 25);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Booking VALUES
    -> (1001, 1, 101, '2026-11-05', 2, 1000, 'Confirmed', 'QR1001'),
    -> (1002, 2, 102, '2026-11-06', 1, 400, 'Confirmed', 'QR1002'),
    -> (1003, 3, 103, '2026-11-07', 1, 700, 'Pending', 'QR1003');
Query OK, 3 rows affected (0.06 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Bus;
+--------+-------------+-----------+-------------+--------+-----------------+
| bus_id | bus_name    | source    | destination | fare   | available_seats |
+--------+-------------+-----------+-------------+--------+-----------------+
|    101 | Express Bus | Chennai   | Bangalore   | 500.00 |              40 |
|    102 | City Travels | Chennai  | Coimbatore  | 400.00 |              30 |
|    103 | Super Deluxe | Bangalore | Hyderabad   | 700.00 |              25 |
+--------+-------------+-----------+-------------+--------+-----------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Bus
    -> WHERE source = 'Chennai'
    -> AND destination = 'Bangalore';
+--------+-------------+---------+-------------+--------+-----------------+
| bus_id | bus_name    | source  | destination | fare   | available_seats |
+--------+-------------+---------+-------------+--------+-----------------+
|    101 | Express Bus | Chennai | Bangalore   | 500.00 |              40 |
+--------+-------------+---------+-------------+--------+-----------------+
1 row in set (0.00 sec)

mysql> SELECT p.passenger_name, b.bus_name,
    ->        b.source, b.destination,
    ->        bk.journey_date, bk.seats,
    ->        bk.total_fare, bk.booking_status
    -> FROM Passenger p
    -> JOIN Booking bk ON p.passenger_id = bk.passenger_id
    -> JOIN Bus b ON bk.bus_id = b.bus_id;
+---------------+-------------+-----------+-------------+--------------+-------+------------+----------------+
| passenger_name | bus_name    | source    | destination | journey_date | seats | total_fare | booking_status |
+---------------+-------------+-----------+-------------+--------------+-------+------------+----------------+
| Ravi          | Express Bus | Chennai   | Bangalore   | 2026-11-05   |     2 |    1000.00 | Confirmed      |
| Priya         | City Travels | Chennai  | Coimbatore  | 2026-11-06   |     1 |     400.00 | Confirmed      |
```

```
mysql> UPDATE Booking bk
    -> JOIN Bus b ON bk.bus_id = b.bus_id
    -> SET bk.total_fare = bk.seats * b.fare;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 3  Changed: 0  Warnings: 0

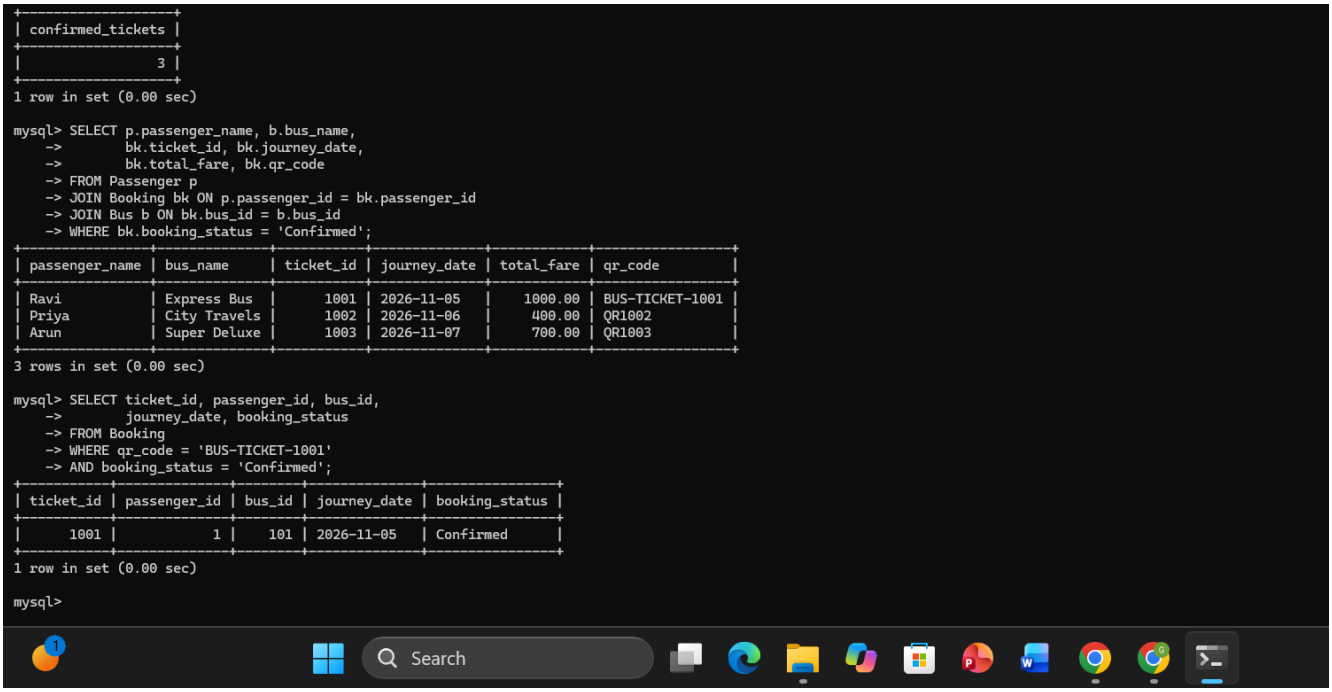
mysql> UPDATE Booking
    -> SET qr_code = CONCAT('BUS-TICKET-', ticket_id)
    -> WHERE ticket_id = 1001;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE Booking
    -> SET booking_status = 'Confirmed'
    -> WHERE ticket_id = 1003;
Query OK, 1 row affected (0.04 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM Booking
    -> WHERE booking_status = 'Confirmed';
+-----------+--------------+--------+--------------+-------+------------+----------------+-----------------+
| ticket_id | passenger_id | bus_id | journey_date | seats | total_fare | booking_status | qr_code         |
+-----------+--------------+--------+--------------+-------+------------+----------------+-----------------+
|      1001 |            1 |    101 | 2026-11-05   |     2 |    1000.00 | Confirmed      | BUS-TICKET-1001 |
|      1002 |            2 |    102 | 2026-11-06   |     1 |     400.00 | Confirmed      | QR1002          |
|      1003 |            3 |    103 | 2026-11-07   |     1 |     700.00 | Confirmed      | QR1003          |
+-----------+--------------+--------+--------------+-------+------------+----------------+-----------------+
3 rows in set (0.00 sec)

mysql> SELECT COUNT(*) AS confirmed_tickets
    -> FROM Booking
    -> WHERE booking_status = 'Confirmed';
+-------------------+
| confirmed_tickets |
+-------------------+
|                 3 |
+-------------------+
1 row in set (0.00 sec)

mysql> SELECT p.passenger_name, b.bus_name,
    ->        bk.ticket_id, bk.journey_date,
    ->        bk.total_fare, bk.qr_code
    -> FROM Passenger p
    -> JOIN Booking bk ON p.passenger_id = bk.passenger_id
    -> JOIN Bus b ON bk.bus_id = b.bus_id
    -> WHERE bk.booking_status = 'Confirmed';
+---------------+-------------+-----------+--------------+------------+-----------------+
| passenger_name | bus_name    | ticket_id | journey_date | total_fare | qr_code         |
```



Observation

The queries manage bus and passenger data, bookings, fare calculations, booking status, and QR references.

Result

Thus, MySQL queries for a QR-enabled bus ticket booking system were implemented successfully.

Note

The QR reference is not an actual QR image. A backend application must generate scannable QR images, validate scans, and manage seat availability.